

Software Requirements Specification (SRS) for Pharmacy Management System

Version 1.0

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1. Introduction

1.1 Purpose

The Pharmacy Management System (PMS) Version 1.0 is a web-based application designed to automate and streamline pharmacy operations. This document specifies the software requirements for the system, covering its functional, non-functional, and external interface requirements.

The PMS is developed using Node.js, MySQL, HTML, CSS, and JavaScript, with an additional machine learning module to analyze sales trends and predict potential disease outbreaks based on medicine demand.

This SRS provides a complete description of the entire system, including its user management, inventory control, sales processing, prescription tracking, supplier management, and machine learning-based analytics. It ensures that pharmacists and administrators can efficiently manage stock, fulfill prescriptions, handle customer transactions, and gain valuable insights into potential disease patterns.

1.2 Document Conventions

This Software Requirements Specification (SRS) follows industry-standard documentation guidelines to ensure clarity, consistency, and ease of reference. The following conventions have been applied throughout this document:

1. Text Formatting:
 - Bold Text – Used for section headings, key terms, and critical concepts.
 - *Italic Text* – Used for emphasis on important points.
 - Monospace Text – Used for code snippets, database field names, and system variables.
2. Requirement Identification & Numbering:
 - FR-x.y – Functional Requirements (e.g., FR-1.2: User Login Authentication).
 - NFR-x.y – Non-functional Requirements (e.g., NFR-2.1: System Response Time).
3. Diagram & Table Labeling:
 - Figures are labeled as Figure X.Y (e.g., *Figure 2.1: System Architecture*).
 - Tables are labeled as Table X.Y (e.g., *Table 3.1: User Role Permissions*).

1.3 Intended Audience and Reading Suggestions

This Software Requirements Specification (SRS) document is intended for individuals involved in the development, evaluation, and usage of the Pharmacy Management System (PMS) as part of a college project. It outlines the system's requirements, functionalities, and constraints, ensuring a structured approach to its implementation.

Intended Audience:

- Project Team (Students) – To understand system requirements, design, and implementation guidelines.

- Faculty & Evaluators – To assess the project’s completeness, technical accuracy, and adherence to software development principles.
- Developers (Students in the Team) – To use this document as a reference for building the system, including database design, front-end, and back-end implementation.
- Testers (Students responsible for Testing) – To verify that the system meets the specified requirements and functions correctly.
- End Users (Pharmacy Staff in Simulation) – To understand how the system will be used for inventory management, sales tracking, and prescription handling.

Reading Suggestions:

To ensure a structured understanding, the document should be read in the following order:

1. Introduction (Section 1) – Overview of the system, including purpose, scope, and objectives.
2. Overall Description (Section 2) – System functionality, user roles, and operational environment.
3. External Interface Requirements (Section 3) – How the system interacts with users, hardware, and software.
4. System Features (Section 4) – Core features like inventory management, sales processing, and predictive analytics.
5. Nonfunctional Requirements (Section 5) – Performance, security, and reliability standards.
6. Additional Sections (Appendices, Section 6 and beyond) – Technical references, diagrams, and analysis models.

2. Overall Description

2.1 Product Perspective

The Pharmacy Management System (PMS) is a new, self-contained application designed to replace manual pharmacy operations. It integrates inventory tracking, prescription management, sales processing, and machine learning analytics to improve efficiency. The system functions independently but can be extended to integrate with larger healthcare management systems.

2.2 Product Functions

- User Management (Admin, Staff, Customers)
- Inventory Management (Stock updates, Expiry tracking)
- Sales & Billing (Invoicing, Payment processing)
- Prescription Handling (Doctor-prescribed medicines)
- Supplier Management (Orders, Deliveries)
- Machine Learning Predictions (Disease outbreak analysis)
- Report Generation (Sales, Inventory trends)

2.3 User Classes and Characteristics

- Admin – Full system access.
- Staff – Sales, inventory management.
- Customers – Purchase & prescription tracking.

2.4 Operating Environment

- Platform: Windows/Linux/macOS
- Database: MySQL
- Backend: Node.js

2.5 Design and Implementation Constraints

- Multi-User Support: The system must support multiple users accessing the system simultaneously without performance degradation.
- Role-Based Authentication: Only authorized users (admins, staff) can access specific functionalities, ensuring security and data protection.
- Database Constraints: The system should handle large volumes of inventory, sales, and customer data without slowdowns.
- Regulatory Compliance: The system must comply with healthcare and data privacy regulations.

2.6 User Documentation

- User Manual: A comprehensive guide covering system usage, inventory management, sales processing, and user access.
- Online Help: Integrated help section for troubleshooting and FAQs accessible within the application.
- Developer Documentation: Technical documentation including API references and database schema for future enhancements.

2.7 Assumptions and Dependencies

- Internet Dependency: The system requires an active internet connection for real-time updates and cloud-based data storage.
- Machine Learning Model Dependency: The accuracy of the disease prediction module depends on historical sales data and external health trends.
- Third-Party Integrations: The system may require integration with third-party APIs for payments, supplier databases, or government health records.
- Hardware Compatibility: The system assumes it will run on standard workstations and mobile devices with a modern web browser.

3. External Interface Requirements

3.1 User Interfaces

The Pharmacy Management System (PMS) will have a web-based graphical user interface (GUI) to facilitate user interaction. The interface will be designed with an intuitive layout, ensuring ease of use for both administrators and staff. The main features of the user interface include:

- Login Page – Secure authentication with username and password.
- Dashboard – Displays an overview of inventory, sales, and prescriptions.
- Inventory Management Page – Allows users to add, update, and track medicines.
- Sales & Billing Page – Enables processing of customer transactions.
- Prescription Management – Allows staff to validate prescriptions and dispense medicines.
- Reports & Analytics – Provides insights into sales trends, low-stock alerts, and ML-based disease predictions.
- Navigation & Standard UI Elements – Consists of buttons, dropdown menus, forms, and modal dialogs for user interactions.

The system will follow standard UI/UX design principles to ensure accessibility, responsiveness, and compatibility across various devices and screen sizes. Error messages and notifications will be displayed in a user-friendly format to guide users through the system.

3.2 Hardware Interfaces

The PMS is designed to run on standard hardware components used in a pharmacy setting. The key hardware interfaces include:

- Workstations & Laptops – The system will be accessible on standard desktop or laptop computers running modern web browsers.
- Barcode Scanner Integration – The system should support barcode scanning devices for quick medicine identification and stock updates.
- Receipt Printers – The system should support thermal receipt printers for generating invoices.
- Cloud Storage – Integration with cloud-based storage solutions for secure data backup.

3.3 Software Interfaces

The PMS will integrate with various software components to ensure smooth operation. Key software interfaces include:

- Database System: MySQL 8.0, used for storing user data, inventory details, transactions, and prescriptions.
- Operating Systems: The system will be compatible with Windows, Linux, and macOS.
- APIs & Libraries: The application will utilize various APIs, including:
 - Flask API for machine learning model integration.
 - Payment Gateway API for online transactions (e.g., Stripe, PayPal).
 - SMS/Email API for sending alerts and notifications.
- Web Technologies: Frontend development will use HTML, CSS, JavaScript (React or Vanilla JS), while the backend will be powered by Node.js with Express.js.

3.4 Communications Interfaces

The PMS will require communication with external and internal components through various network and protocol standards. The key communication interfaces include:

- **Web Browser Communication:** The system will operate over HTTPS for secure data transmission.
- **REST API Communication:** The backend will expose RESTful APIs for frontend communication.
- **Email Notifications:** The system will use SMTP-based email services to send user notifications regarding inventory alerts, order confirmations, and reports.
- **Network Connectivity:** The system will require internet access for cloud-based database management and remote access.
- **Data Security Measures:** All communication will be encrypted using SSL/TLS protocols to protect sensitive pharmacy data.

4. System Features

4.1 Inventory Management

4.1.1 Description and Priority

The Inventory Management feature is a high-priority component, ensuring that medicines are accurately tracked, stocked, and managed. This functionality helps prevent shortages and overstocking, enabling seamless pharmacy operations.

4.1.2 Stimulus/Response Sequences

1. **User Action:** Staff logs into the system and accesses the inventory page.
System Response: Displays the current stock levels of all medicines.
2. **User Action:** Staff updates stock quantity after receiving a new supply.
System Response: The system records the update and logs the changes in the inventory history.
3. **User Action:** The system detects that a medicine is low in stock.
System Response: Sends an alert notification to the administrator.

4.1.3 Functional Requirements

- REQ-4.1.1: The system must allow users to add, update, and remove medicines from inventory.
- REQ-4.1.2: The system must track stock levels and expiry dates of medicines.
- REQ-4.1.3: The system should generate low-stock alerts for medicines falling below a threshold.
- REQ-4.1.4: The system must maintain a history of stock updates for auditing purposes.

4.2 Sales & Billing

4.2.1 Description and Priority

The Sales & Billing module is a high-priority feature that ensures accurate sales transactions and invoice generation. It enables seamless processing of customer purchases and payment handling.

4.2.2 Stimulus/Response Sequences

1. **User Action:** Staff selects medicines and enters the quantity for a customer.
System Response: The system calculates the total price and applies discounts if applicable.
2. **User Action:** Staff processes payment through cash or digital methods.
System Response: The system generates a receipt and updates inventory levels.

4.2.3 Functional Requirements

- REQ-4.2.1: The system must support medicine selection and quantity input for billing.
- REQ-4.2.2: The system must calculate total price, discounts, and taxes.
- REQ-4.2.3: The system must generate an invoice for every completed transaction.
- REQ-4.2.4: The system must update inventory stock after a sale is made.

4.3 Prescription Handling

4.3.1 Description and Priority

The Prescription Handling feature is a medium-priority component that ensures medicines are dispensed only with a valid prescription.

4.3.2 Stimulus/Response Sequences

1. User Action: Staff enters prescription details into the system.
System Response: The system validates the prescription and displays approved medicines.
2. User Action: Staff selects the prescribed medicines for dispensing.
System Response: The system verifies stock availability and updates the database.

4.3.3 Functional Requirements

- REQ-4.3.1: The system must allow prescription verification before dispensing medicines.
- REQ-4.3.2: The system must validate that the prescribed medicine exists in inventory.
- REQ-4.3.3: The system must store prescription records for future reference.

4.4 Machine Learning Predictions

4.4.1 Description and Priority

The Machine Learning Predictions module is a low-priority feature that analyzes medicine sales trends to detect potential disease outbreaks and recommend stock adjustments.

4.4.2 Stimulus/Response Sequences

1. User Action: Admin accesses the machine learning analytics dashboard.
System Response: Displays predicted medicine demand and possible outbreak patterns.
2. User Action: Admin reviews recommended stock adjustments.
System Response: The system provides suggested restocking actions based on ML predictions.

4.4.3 Functional Requirements

- REQ-4.4.1: The system should analyze medicine sales trends using machine learning models.
- REQ-4.4.2: The system must generate alerts if an unusual sales pattern is detected.
- REQ-4.4.3: The system should recommend stock adjustments based on predictions.

This ensures that the system maintains efficiency, security, and intelligent decision-making in pharmacy operations.

4.5 Supplier & Order Management

- Description: Manages supplier details and tracks medicine orders.
- Priority: High
- Functional Requirements:
 - REQ-4.5.1: The system must store and manage supplier details.
 - REQ-4.5.2: The system must allow users to create and track purchase orders.
 - REQ-4.5.3: The system should update inventory when an order is received.

4.6 User Management & Role-Based Access

- Description: Ensures secure user authentication and access control.
- Priority: High
- Functional Requirements:
 - REQ-4.6.1: The system must support admin and staff roles with different permissions.
 - REQ-4.6.2: The system should enforce secure login authentication.
 - REQ-4.6.3: The system should allow admins to manage user accounts.

4.7 Reporting & Analytics

- Description: Generates reports on sales, inventory, and ML predictions.
- Priority: Medium
- Functional Requirements:
 - REQ-4.7.1: The system should generate sales reports with revenue analytics.
 - REQ-4.7.2: The system must provide inventory usage and stock alerts.
 - REQ-4.7.3: The system must display machine learning insights on demand trends.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

- The system should be able to handle simultaneous access by multiple users without performance degradation.
- Inventory searches should return results in under 2 seconds.
- Sales transactions should be processed and recorded in less than 3 seconds.
- The system should support at least 100,000 records in the database without significant performance issues.
- The machine learning module should analyze sales data and generate predictions within 5 minutes after a major transaction update.

5.2 Safety Requirements

- The system should prevent the sale of expired medicines by automatically flagging them as unavailable.
- Data integrity must be maintained to prevent accidental deletion of critical records such as prescriptions, patient history, and sales transactions.
- The system should provide automatic daily backups to prevent data loss.
- The software must comply with health and safety regulations for pharmacy operations to ensure legal compliance.

5.3 Security Requirements

- The system should implement role-based access control (RBAC) to restrict access to sensitive information.
- Passwords should be hashed and stored securely using industry-standard encryption methods (e.g., bcrypt or SHA-256).
- Two-factor authentication (2FA) should be implemented for administrator accounts.
- User activity logs should be maintained to track all important system interactions.
- The system should use SSL/TLS encryption for all data transfers between clients and servers.

5.4 Software Quality Attributes

- **Availability:** The system should maintain 99.9% uptime, with scheduled maintenance performed during non-peak hours.
- **Reliability:** The system should ensure that critical functions, such as inventory management and sales processing, operate without failures.
- **Usability:** The UI should be intuitive and user-friendly, allowing new users to learn its functions in under 30 minutes.
- **Scalability:** The system should be designed to support expansion, allowing additional pharmacies or users to be added without significant modifications.
- **Maintainability:** The system should be modular, making it easy to update specific components without affecting overall functionality.

5.5 Business Rules

- Only authorized staff can process sales transactions and update inventory records.
- Prescription-based medicines cannot be sold without valid doctor authorization.
- The system must maintain records of all purchases and prescriptions for regulatory compliance.
- Discounts and promotional offers should be configurable only by administrators.
- Suppliers must be verified and approved before they can be added to the system for order fulfillment.

6. Other Requirements

6.1 Database Requirements

- The system must use MySQL 8.0 as the relational database management system (RDBMS).
- All transactions should follow ACID (Atomicity, Consistency, Isolation, Durability) properties to ensure data integrity.
- The database should support automated daily backups to prevent data loss.
- Data archiving mechanisms must be implemented to store historical sales and prescription data for analysis.

6.2 Internationalization Requirements

- The system should support multi-language capabilities, allowing users to switch between English and other regional languages.
- Currency formats should be configurable based on the pharmacy's location.
- Date and time formats must be adaptable to different regions.

6.3 Legal and Compliance Requirements

- The system must comply with GDPR (General Data Protection Regulation) and HIPAA (Health Insurance Portability and Accountability Act) regulations for data privacy and protection.
- Pharmaceutical compliance must be maintained by ensuring that prescription-based medicines cannot be sold without valid authorization.
- The system must log all transactions for audit and regulatory compliance.

6.4 Reusability and Extensibility

- The system should be designed using modular architecture, allowing future enhancements without affecting core functionality.
- APIs should be well-documented to facilitate integration with third-party applications such as insurance providers and hospital management systems.
- The machine learning model should be adaptable, allowing future training with new datasets to improve prediction accuracy.

6.5 Environmental & Hardware Constraints

- The system should run on Windows, Linux, and macOS operating systems.
- The minimum system requirements include:
 - Processor: Intel Core i3 or higher
 - RAM: 8GB or more
 - Storage: 50GB of available disk space
 - Network: Stable internet connection for cloud-based features

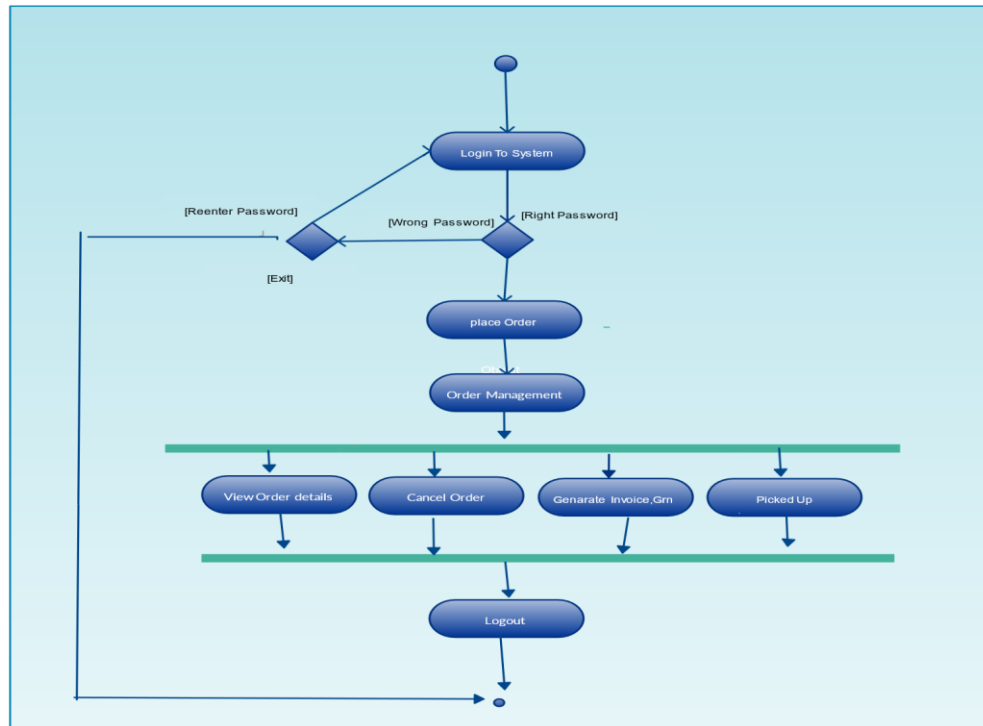
Appendix A: Glossary

This section defines key terms, acronyms, and abbreviations used in this document to ensure clarity and understanding.

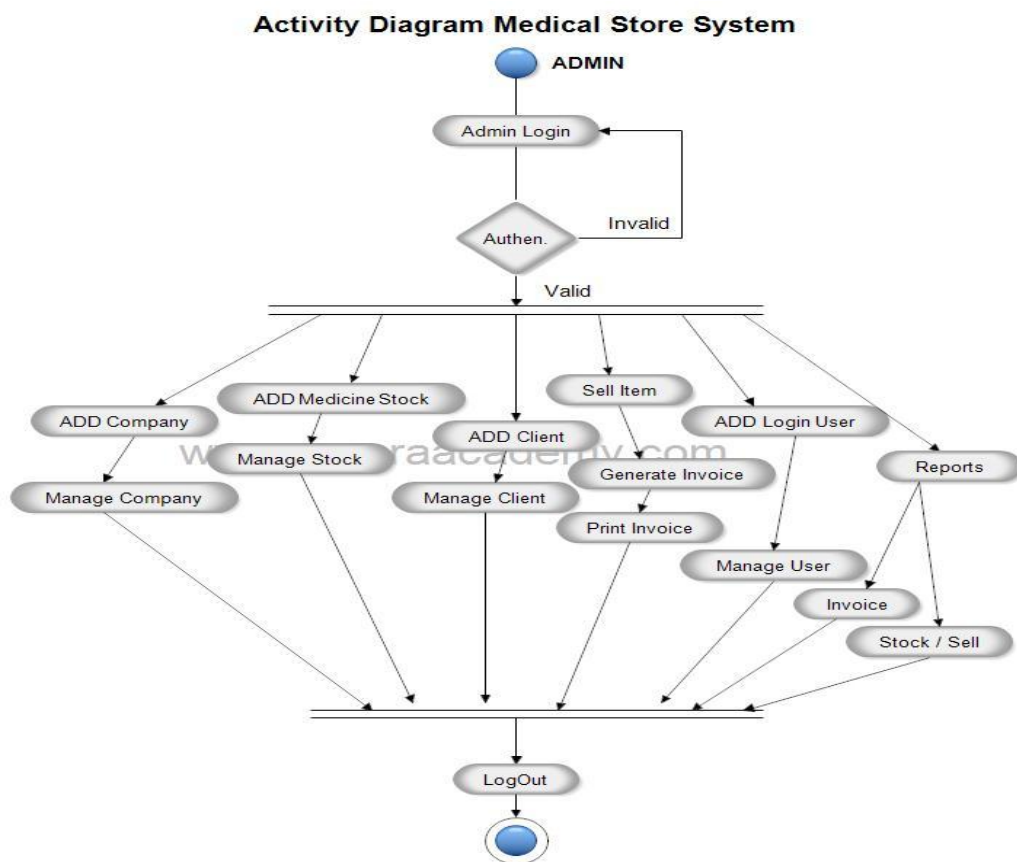
- **PMS (Pharmacy Management System):** The web-based application designed to manage pharmacy operations.
- **RBAC (Role-Based Access Control):** A security model ensuring that users have access only to the functionalities assigned to their roles.
- **ACID (Atomicity, Consistency, Isolation, Durability):** A set of database properties ensuring reliable processing of transactions.
- **HIPAA (Health Insurance Portability and Accountability Act):** A U.S. regulation ensuring the security and privacy of health-related data.
- **GDPR (General Data Protection Regulation):** A regulation governing data protection and privacy in the European Union.
- **SSL/TLS (Secure Sockets Layer/Transport Layer Security):** Encryption protocols used to secure data transmission over networks.
- **2FA (Two-Factor Authentication):** An additional security measure requiring users to verify their identity through two different authentication factors.
- **API (Application Programming Interface):** A set of protocols for building and integrating software applications.
- **SKU (Stock Keeping Unit):** A unique identifier assigned to each type of item in inventory for tracking purposes.
- **Machine Learning (ML):** A type of artificial intelligence that enables the system to analyze data and make predictions based on patterns.

- ER Diagram of Pharmacy Management System:

- Pharmacy Order Processing Flowchart



- Activity Diagram



This section includes a list of elements within the Pharmacy Management System that require further clarification, research, or finalization before project completion. These items will be tracked and resolved as development progresses.

TBD Items and Their Status

1. Choice of Machine Learning Algorithm for Disease Prediction – TBD: Decision pending between Random Forest, SVM, or Neural Networks.
2. Integration of Payment Gateway – TBD: Selection between PayPal, Stripe, or Razorpay.
3. Multi-Language Support Implementation – TBD: Confirmation of supported languages and translation method.
4. Security Measures for Data Encryption – TBD: Choice between AES-256 or RSA Encryption for sensitive data storage.
5. Cloud Hosting vs. On-Premise Deployment – TBD: Decision on whether the system will be self-hosted or cloud-based (AWS, Azure, or GCP).
6. User Access Levels and Permissions – TBD: Finalization of detailed role-based access for admins, staff, and suppliers.
7. Report Generation Format (PDF, Excel, or Web Dashboard) – TBD: Decision on the preferred report format for pharmacy insights and analytics.
8. Inventory Stock Threshold Levels – TBD: Determination of minimum stock levels to trigger automatic restocking alerts.

HALOmed Pharmacy Management System – Database Normalization

This document outlines the normalization process for each table in the HALOmed Pharmacy Management System. It includes the functional dependencies and step-by-step analysis to reach the final normal form.

1. Users Table

Attributes: `user_id`, `username`, `password_hash`, `role`

Functional Dependencies:

- `user_id` → `username`, `password_hash`, `role`
- `username` → `user_id`, `password_hash`, `role` (if unique)

Normalization Process:

- 1NF: All attributes are atomic ✓
- 2NF: No partial dependencies ✓
- 3NF: No transitive dependencies ✓
- BCNF: Every determinant is a candidate key ✓

Final Normal Form: BCNF

2. Customers Table

Attributes: `customer_id`, `name`, `contact`, `address`

Functional Dependencies:

- `customer_id` → `name`, `contact`, `address`
- `contact` → `customer_id`, `name`, `address` (if unique)

Normalization Process:

- 1NF: All attributes are atomic ✓
- 2NF: No partial dependencies ✓
- 3NF: No transitive dependencies ✓

- BCNF: Every determinant is a candidate key ✓

Final Normal Form: BCNF

3. Medicines & MedicineBatch Tables

Medicines Attributes: `medicine_id`, `name`, `category`

MedicineBatch Attributes: `batch_id`, `medicine_id`, `expiry_date`

Functional Dependencies:

- `medicine_id` → `name`, `category`
- `batch_id` → `medicine_id`, `expiry_date`

Normalization Process:

- Original table failed 2NF due to dependency on batches
- Decomposed into Medicines and MedicineBatch
- All resulting tables meet BCNF

Final Normal Form: BCNF

4. Suppliers Table

Attributes: `supplier_id`, `name`, `contact`, `address`

Functional Dependencies:

- `supplier_id` → `name`, `contact`, `address`
- `contact` → `supplier_id`, `name`, `address` (if unique)

Normalization Process:

- All normal forms satisfied

Final Normal Form: BCNF

5. Orders & OrderDetails Tables

Orders Attributes: `order_id`, `supplier_id`, `order_date`

OrderDetails Attributes: `order_id`, `medicine_id`, `quantity`

Functional Dependencies:

- `order_id` → `supplier_id`, `order_date`
- `order_id`, `medicine_id` → `quantity`

Normalization Process:

- Original Orders table failed 2NF due to derived total
- Decomposed into Orders and OrderDetails
- Resulting tables satisfy BCNF

Final Normal Form: BCNF

6. InventoryLogs Table

Attributes: `log_id`, `medicine_id`, `change_type`, `quantity_changed`, `timestamp`

Functional Dependencies:

- `log_id` → all attributes

Final Normal Form: BCNF

7. Sales & SaleDetails Tables

Sales Attributes: `sale_id`, `customer_id`, `sale_date`

SaleDetails Attributes: `sale_id`, `medicine_id`, `quantity`, `price`

Functional Dependencies:

- `sale_id` → `customer_id`, `sale_date`
- `sale_id`, `medicine_id` → `quantity`, `price`

Normalization Process:

- Price was dependent on medicine not sale
- Decomposed to meet BCNF

Final Normal Form: BCNF

8. Prescriptions Table

Attributes: prescription_id, customer_id, medicine_id, prescribed_by, date_issued

Functional Dependencies:

- prescription_id → all other attributes

Final Normal Form: BCNF

9. PaymentTransactions Table

Attributes: transaction_id, sale_id, payment_method, amount_paid, payment_date

Functional Dependencies:

- transaction_id → all other attributes

Final Normal Form: BCNF

10. DiseaseOutbreakPredictions Table

Attributes: prediction_id, predicted_disease, confidence_score, prediction_date

Functional Dependencies:

- prediction_id → all other attributes

Final Normal Form: BCNF

11. MedicineRecommendations Table

Attributes: recommendation_id, prediction_id, medicine_id, recommendation_reason

Functional Dependencies:

- recommendation_id → all other attributes

Final Normal Form: BCNF

12. DemandAnalytics Table

Attributes: analytics_id, medicine_id, total_sales, avg_daily_demand, last_updated

Functional Dependencies:

- analytics_id → all other attributes

Final Normal Form: BCNF

13. Reports Table

Attributes: report_id, report_type, generated_date, report_content

Functional Dependencies:

- report_id → all other attributes

Final Normal Form: BCNF



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MANIPAL
A Constituent Unit of MAHE, Manipal

MEDHUB: Pharmacy Management System

A Project Report Submitted

to

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ABSTRACT

This project presents MEDHUB, a comprehensive Pharmacy Management System designed to streamline the operations of pharmacies by automating sales, inventory, prescription handling, and supplier order management. The system features user authentication and role-based access to ensure secure data handling and accountability. Through normalized relational database design, MEDHUB ensures efficient data integrity and scalability. Core modules include inventory tracking, real-time stock updates, customer sales records, prescription management, supplier orders, and advanced reporting functionalities. Integrated support for predictive analytics enables disease outbreak forecasts and medicine recommendations based on historical data, promoting proactive healthcare practices. Designed with modular architecture, MEDHUB is adaptable for use in pharmacies of varying scales and serves as a robust foundation for further integration with AI and IoT-based systems.

Keywords:

Pharmacy management system; Relational databases; Inventory control; Prescription tracking; Health informatics; Disease prediction; Sales management; Supplier integration; Report generation; Decision support systems; Healthcare automation; Data-driven recommendations; Sustainable healthcare IT.

Sustainable Development Goal:

3. Good Health and Well-being

MEDHUB supports Sustainable Development Goal 3 by promoting health and well-being for all through improved access to essential medicines, reduced medication errors, and enhanced pharmacy service delivery using digital technology.

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Chapter 1 Introduction

The pharmaceutical industry plays a crucial role in ensuring access to healthcare and the timely availability of essential medicines. However, the management of inventory, prescriptions, customer data, and supplier coordination in pharmacies remains a challenge due to manual processes and outdated systems. MEDHUB addresses these issues through a streamlined, webbased Pharmacy Management System that integrates all essential functionalities into a single, efficient platform.

1.1 Background and Significance:

Traditional pharmacy operations often rely on paper-based systems or basic digital tools with limited integration. These fragmented solutions lead to inefficiencies such as inventory mismatches, sale of expired medicines, slow billing, and poor customer record maintenance. In addition, there is a lack of predictive mechanisms to forecast medicine demand based on historical data or health trends.

MEDHUB bridges these gaps by implementing a robust, relational database-driven solution that integrates inventory control, sales processing, prescription validation, and analytics. It also introduces machine learning-based predictions to aid in stock planning and disease outbreak preparedness. This comprehensive approach improves operational efficiency and supports better healthcare delivery.

1.2 Need for Computerized Pharmacy Management:

Pharmacies face several limitations with conventional systems, which MEDHUB aims to resolve:

1. **Data Silos:** Sales, inventory, and prescriptions are often managed in isolation, reducing operational clarity.
2. **Human Error:** Manual updates can result in data inaccuracies or delays, especially during high traffic periods.
3. **Limited Insights:** Traditional systems lack analytical tools for sales forecasting or outbreak detection.
4. **Stock Mismanagement:** Medicines may be understocked or overstocked due to poor inventory visibility.
5. **Lack of Integration:** No linkage between prescriptions, sales, and recommendations.

A computerized system like MEDHUB provides automation, improves decision-making, and ensures real-time access to pharmaceutical data. It empowers pharmacists to manage medicines efficiently and make proactive business and healthcare decisions.

1.3 Overview of the Proposed System:

MEDHUB follows a modular architecture consisting of:

1. **Presentation Layer:** A web interface for pharmacists, staff, and administrators, providing dashboards for sales, stock, prescriptions, and analytics.
2. **Application Layer:** Business logic modules to process transactions, manage inventory, track prescriptions, and generate alerts or recommendations.
3. **Data Layer:** A structured MySQL-based database that stores users, medicines, prescriptions, customers, suppliers, transactions, and ML predictions.

MEDHUB manages all key pharmacy functions in an integrated environment. Entities include Users, Medicines, Sales, InventoryLogs, Prescriptions, Suppliers, Orders, PaymentTransactions, and DiseaseOutbreakPredictions. The system ensures normalization (BCNF), data integrity, and fast querying for scalable use.

1.4 Project Objectives:

The primary objectives of MEDHUB are to:

1. Design and implement a relational database to manage pharmacy operations.
2. Enable end-to-end inventory management with expiry alerts and low-stock notifications.
3. Integrate prescription validation and sales processing for better compliance.
4. Implement a machine learning model to forecast disease outbreaks based on medicine sales.
5. Provide detailed reporting and analytics to administrators.
6. Support a scalable and secure framework with multi-user access.

By achieving these goals, MEDHUB supports good health and well-being through efficient medicine distribution and intelligent pharmacy operation management.

Chapter 2 Literature Survey and Background

2.1 Background of Pharmacy Management Systems:

Pharmacy management systems are essential components of healthcare infrastructure. Traditionally, pharmacies used manual ledger systems to record sales and maintain stock. With the advent of digital technology, these operations have moved to computerized systems, improving accuracy, transparency, and efficiency. In developed countries, integrated pharmacy systems support real-time prescription validation and centralized databases. However, many pharmacies in developing countries still struggle with outdated or isolated systems.

The need for a comprehensive and scalable solution has led to the development of modern web-based systems like MEDHUB. These systems combine relational database architecture, cloud compatibility, and machine learning capabilities to deliver intelligent and responsive solutions.

2.2 Review of Existing Systems:

2.2.1 Global Systems:

1. Cerner Pharmacy: Provides complete hospital pharmacy integration with real-time clinical data access.
2. Epic Willow: Used in the U.S. for medication tracking across inpatient and outpatient settings with safety checks and alerts.
3. McKesson Connect: Offers inventory control and order management for chain pharmacies.

2.2.2 Indian Context

In India, many local pharmacies use basic POS software with limited connectivity or data integration. Cloud-based platforms like Pharमारack and MedCords are emerging to fill the gap by offering online ordering, medicine delivery, and digital prescription storage. However, comprehensive pharmacy ERP systems with ML-driven analytics are still uncommon. MEDHUB stands apart by incorporating ML-based demand forecasting and outbreak prediction in a user-friendly format for academic and professional use.

2.3 Technological Approaches in Pharmacy Systems

2.3.1 Database Architectures

MEDHUB uses a normalized MySQL database that ensures ACID compliance, fast transaction processing, and secure storage. Each table is designed in BCNF, reducing redundancy and improving query performance. The system supports dynamic joins and scalable inserts for thousands of records without significant lag.

2.3.2 Machine Learning Integration

Unlike traditional systems, MEDHUB integrates machine learning for disease outbreak forecasting. It analyzes sales trends to predict spikes in demand for specific medicines, which may indicate early signs of local health issues. This predictive capability helps pharmacies prepare stock accordingly, potentially aiding public health efforts.

2.3.3 Web Technologies

MEDHUB is built using Node.js and Express.js on the backend, with frontend support via HTML, CSS, and optionally React or Vue.js. The ML component is implemented using Flask in Python, communicating through REST APIs. This modern stack allows modular development, faster loading times, and easier third-party integration.

2.4 Research Gaps Addressed by This Project

1. Lack of integrated analytics in traditional pharmacy systems.
2. Minimal prescription recommendation logic based on historical data.
3. Poor inventory prediction and demand forecasting mechanisms.
4. Absence of role-based access and audit trails in existing lightweight apps.
5. Technological Trends
6. Predictive Inventory Management using ML
7. Cloud-hosted backups and data synchronization
8. RESTful APIs for cross-platform integration
9. Real-time dashboards with data visualization

Chapter 3 Methodology

3.1 Problem Statement

Pharmacies face numerous operational challenges, including inaccurate stock management, outdated prescription handling, lack of transaction traceability, and no mechanisms to anticipate medicine demand. These issues result in medicine shortages, delayed service, and inefficient supplier coordination. MEDHUB aims to solve these issues by offering a modular, data-driven platform.

3.2 Project Objectives

1. To design and normalize a relational database for pharmacy operations.
2. To implement modules for inventory, billing, customer and supplier management.
3. To enable secure login with role-based access control.
4. To incorporate ML predictions for outbreak trends and recommendations.
5. To provide automated reporting and real-time alerts.

3.3 Scope of the Project

The scope includes the creation of a web-based system for managing pharmacy transactions, inventory, prescriptions, supplier orders, and report generation. The project includes database normalization up to BCNF, machine learning integration, and responsive UI development. It excludes mobile app development and hardware integrations like barcode scanners for this version.

Chapter 4 Data

Design

4.1 Database Schema

The MEDHUB system's relational database consists of multiple well-structured primary tables and junction tables. These tables manage medicine inventory, customer transactions, and intelligent prediction integration:

1. USERS

- Primary Key: user_id
- Attributes: username, password_hash, role, created_at
- Purpose: Stores credentials and roles of administrative and operational users.

2. CUSTOMERS

- Primary Key: customer_id
- Attributes: name, contact_info, address
- Purpose: Stores client/patient details for tracking prescriptions and sales.

3. MEDICINES

- Primary Key: medicine_id
- Attributes: name, category, price, stock_quantity, expiry_date
- Purpose: Maintains inventory details, classifications, pricing, and expiration data.

4. SUPLIERS

- Primary Key: supplier_id
- Attributes: name, contact_info, address
- Purpose: Stores information about medicine suppliers.

5. ORDERS

- Primary Key: order_id
- Foreign Keys: supplier_id (Suppliers), created_by (Users)
- Attributes: order_date, total_amount
- Purpose: Tracks medicine orders placed with suppliers.

6. INVENTORYLOGS

- Primary Key: log_id
- Foreign Keys: medicine_id (Medicines), changed_by (Users)
- Attributes: quantity_changed, change_type, log_date
- Purpose: Maintains records of stock updates for audit purposes.

7. SALES

- Primary Key: sale_id
- Foreign Key: customer_id (Customers)
- Attributes: sale_date
- Purpose: Records each customer transaction.

8. SALEDETAILS

- Primary Key: sale_detail_id
- Foreign Keys: sale_id (Sales), medicine_id (Medicines)
- Attributes: quantity, price
- Purpose: Holds the line items for each sale.

9. PRESCRIPTIONS

- Primary Key: prescription_id
- Foreign Keys: customer_id (Customers), medicine_id (Medicines)
- Attributes: prescribed_by, date_issued
- Purpose: Records medicine prescriptions issued to customers.

10. PAYMENTTRANSACTIONS

- Primary Key: transaction_id
- Foreign Key: sale_id (Sales)
- Attributes: payment_method, amount_paid, payment_date
- Purpose: Manages payments made for medicine purchases.

11.DISEASEOUTBREAKPREDICTION

- Primary Key: prediction_id
- Attributes: predicted_disease, confidence_score, prediction_date
- Purpose: Stores machine learning-based forecasts of disease outbreaks.

12.MEDICINERECOMMENDATION

- Primary Key: recommendation_id
- Foreign Keys: prediction_id (DiseaseOutbreakPredictions), medicine_id (Medicines)
 - Attributes: recommendation_reason
- Purpose: Associates medicines with predicted diseases and reasons for recommendation.

13.REPORTS

- Primary Key: report_id
- Foreign Key: generated_by (Users)
- Attributes: report_type, content, report_date
- Purpose: Generates summaries and analytics based on system activities.

4.2 Entity-Relationship Model

The ER diagram depicts conceptual data organization through:

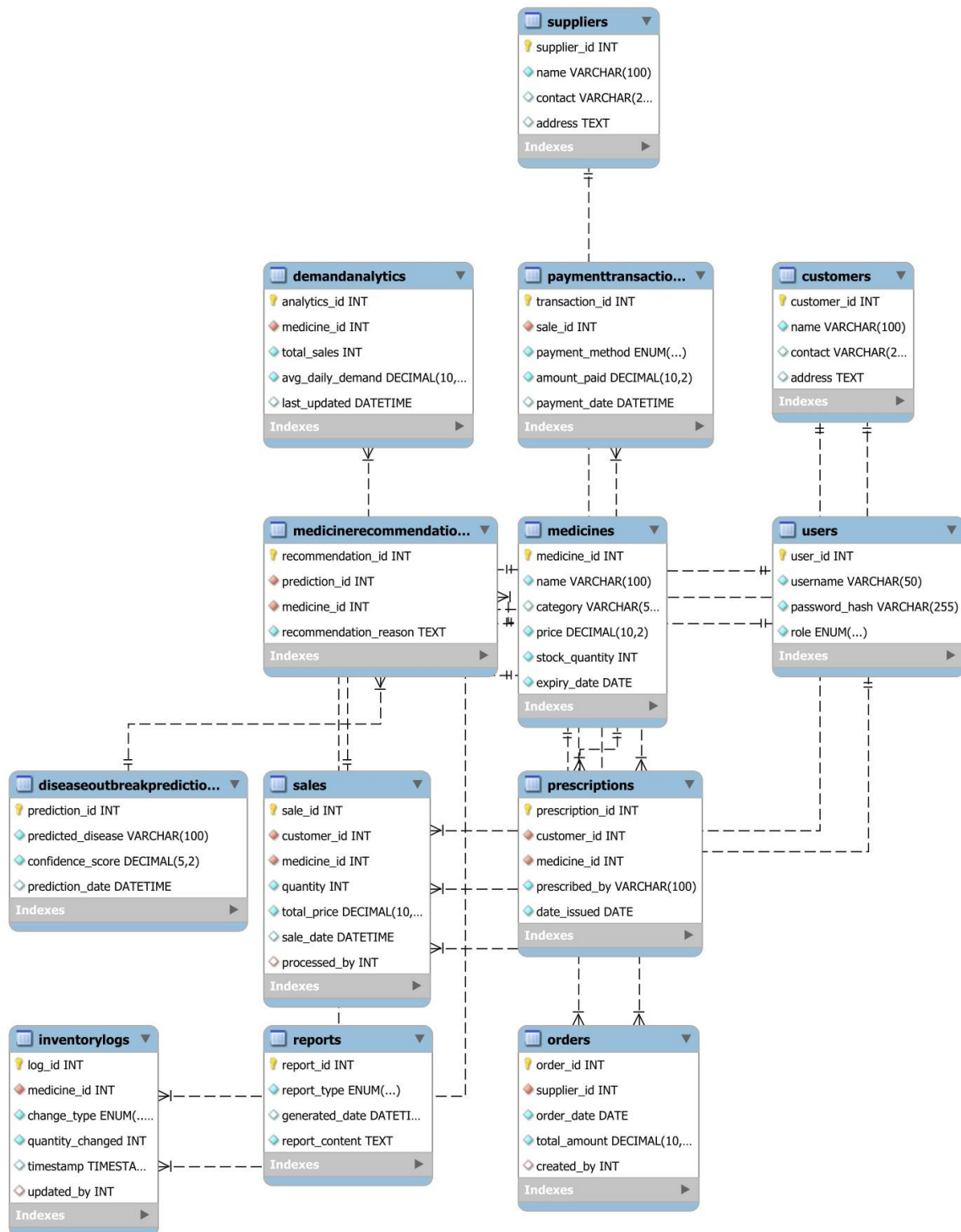


Figure 4.1: Entity Relation Diagram of MEDHUB Pharmacy Management System

1. Entities (Rectangles):

- Primary entities such as Users, Medicines, Customers, Suppliers, Orders, and Sales.
- Each entity contains attributes representing columns in the database.

2. Relationships (Lines):

- Users are linked to Reports, Orders (created_by), and InventoryLogs (changed_by).
- Customers relate to Sales and Prescriptions.
- Medicines are central to InventoryLogs, SaleDetails, Prescriptions, and MedicineRecommendations.
- Sales are linked to PaymentTransactions and SaleDetails.
- DiseaseOutbreakPredictions relate to MedicineRecommendations.

3. Attributes:

- Simple attributes like name, category, price, etc.
- No multivalued or composite attributes were used, ensuring BCNF compliance.

4.3 Normalization Process

The database has been normalized to Boyce-Codd Normal Form (BCNF) through the following steps:

1. 1NF (First Normal Form):
 - Eliminated repeating groups and ensured all attributes are atomic.
 - Defined primary keys for each entity to ensure unique identification.
2. 2NF (Second Normal Form):
 - Removed partial dependencies by decomposing tables where non-key attributes did not depend on full primary keys.
 - Introduced junction tables like SaleDetails and MedicineRecommendations.
3. 3NF (Third Normal Form):
 - Removed transitive dependencies by ensuring non-key attributes are not dependent on other non-key attributes.
 - Attributes like medicine category or user role are only dependent on their direct primary keys.
4. BCNF (Boyce-Codd Normal Form):
 - Ensured every determinant is a candidate key.
 - Reorganized relations such as Orders, SaleDetails, and InventoryLogs for high integrity and low redundancy.
 - All tables follow strict foreign key references, enhancing relational integrity.

Users

user_id username password_hash role created_at

Customers

customer_id name contact_info address

Medicines medicine_id name category price stock_quantity expiry_date

Suppliers

supplier_id name contact_info address

Orders

order_id supplier_id order_date total_amount created_by

InventoryLogs

log_id medicine_id quantity_changed change_type log_date changed_by

Sales

sale_id customer_id sale_date

SaleDetails

sale_detail_id sale_id medicine_id quantity price

Cart

cart_item_id customer_id medicine_id quantity unit_price subtotal discounts total_price added_at

Prescriptions

prescription_id customer_id medicine_id prescribed_by date_issued

PaymentTransactions transaction_id sale_id payment_method amount_paid
payment_date

DiseaseOutbreakPredictions prediction_id predicted_disease
confidence_score prediction_date

MedicineRecommendations recommendation_id prediction_id medicine_id
recommendation_reason

Reports

report_id	report_type	generated_by	content	report_date
-----------	-------------	--------------	---------	-------------

Chapter 5 Methodology

5.1 Inventory Data Aggregation Functions :

The system implements three key functions to analyse pharmacy-related data:

1. Stock Level Aggregation: Calculates real-time stock quantities using inventory logs and medicine table joins.
2. Price Analysis: Computes average medicine pricing across categories to support customer budget planning.
3. Expiry Monitoring: Tracks near-expiry medicines using date comparisons to enable timely disposal or discounts.

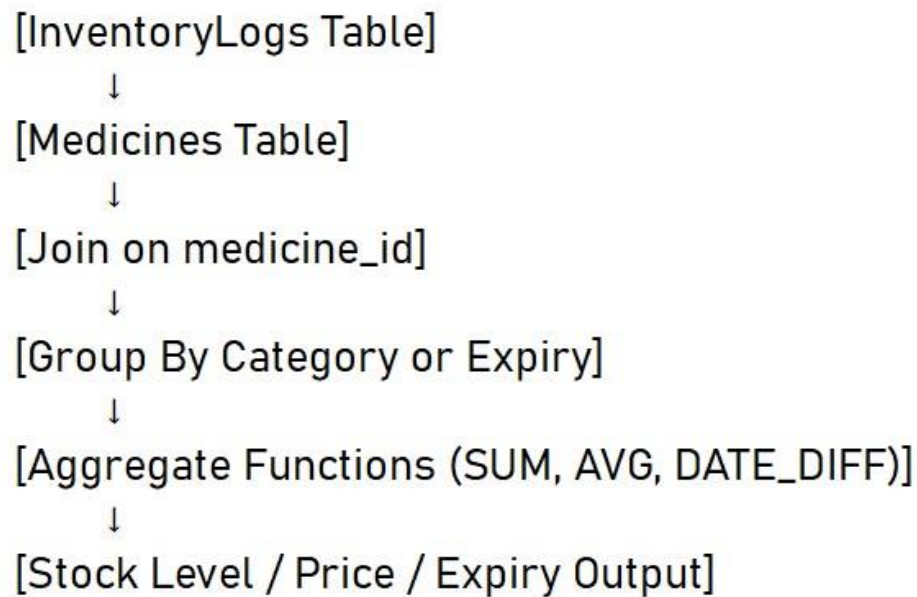


Figure 5.1: Process Flow of Inventory Data Aggregation Function

5.2 Medicine Recommendation Generation Procedure A cursor-based procedure matches disease outbreak predictions with suggested medicines:

1. Prediction-Medicine Mapping: Uses disease names from outbreak predictions to fetch mapped medicines.
2. Dynamic Recommendations: Returns medicines with justifications from the MedicineRecommendations table.
3. Error Handling: Includes exception mechanisms for missing data or null mappings.

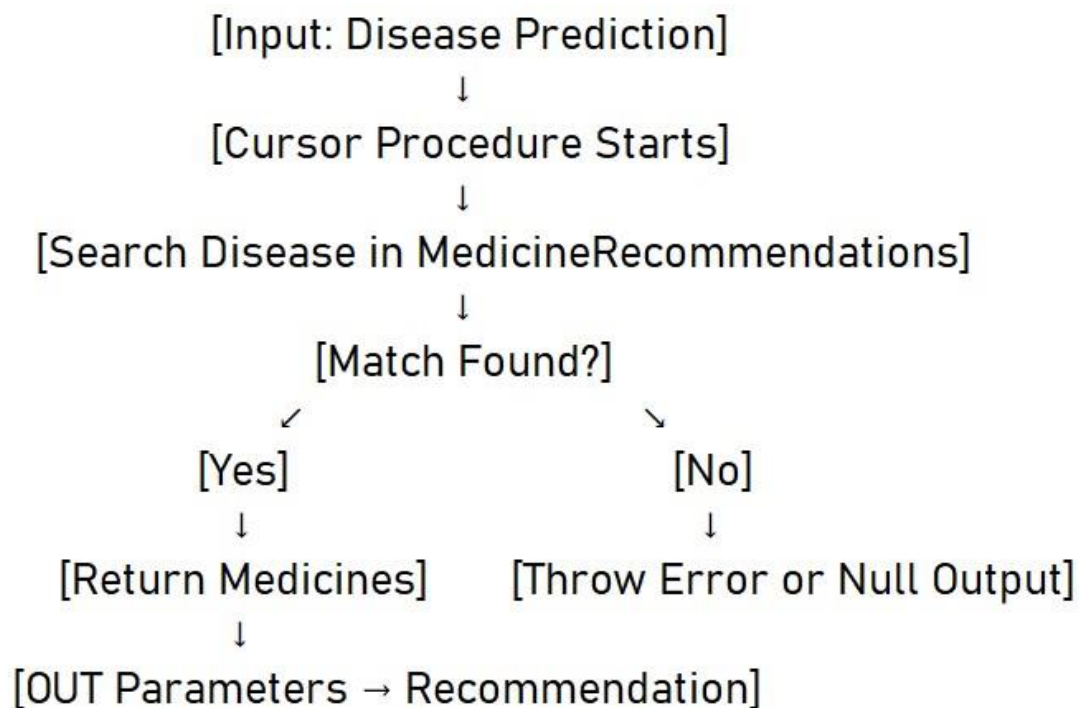


Figure 5.2: Decision Flow of Medicine Recommendation Generation Procedure

5.3 Prescription and Sales Query Analysis Two nested queries offer insights into prescription trends and sales performance:

1. Top Prescribed Medicines: Ranks frequently prescribed items based on recent records.
2. Sales Volume Insight: Aggregates item-wise revenue using joins across SaleDetails, Sales, and Medicines.

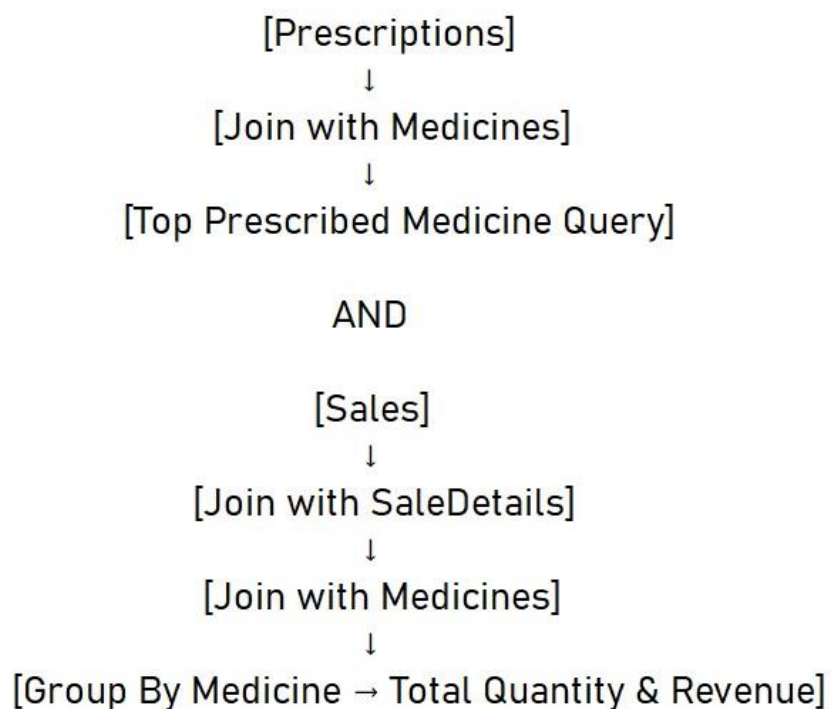
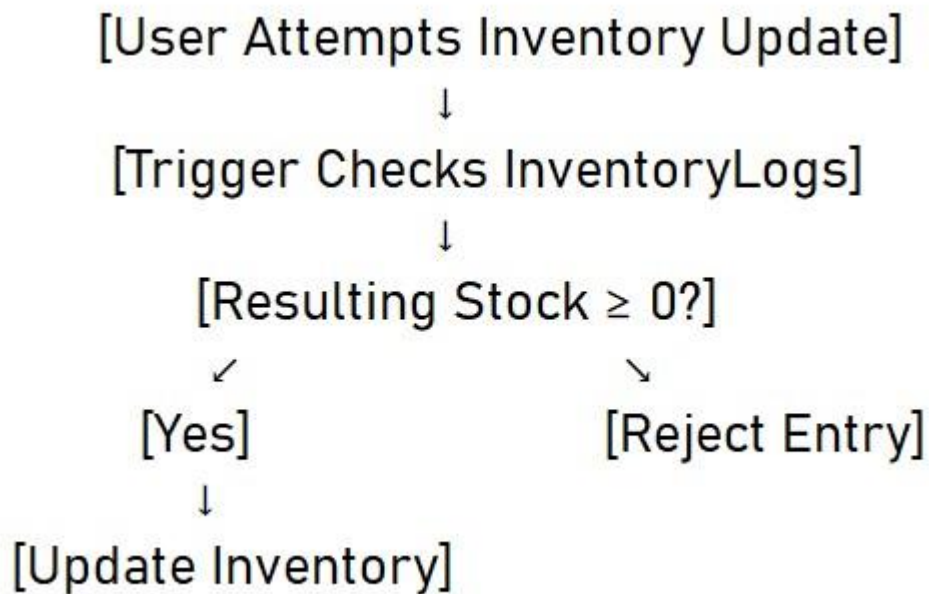


Figure 5.3: Data Flow of Prescription and Sales Queries

5.4 Automated Data Integrity Mechanisms Two triggers ensure data validation and reliability:

1. Negative Stock Prevention: Blocks log entries that would result in negative stock levels.
2. User Creation Notification: Issues confirmation messages when a new user account is created.



Validation Flow of Automated Data Integrity Mechanisms

Figure 5.4:

Chapter 6

Results

The MEDHUB Pharmacy Management System demonstrates successful integration of inventory, prescriptions, billing, and forecasting components. It supports secure logins, intuitive dashboards, and efficient tracking of inventory and sales. Real-time alerts for low-stock and expired medicines enhance operational responsiveness. Predictive analytics modules suggest future demand trends and potential disease outbreaks, empowering proactive stock decisions.

MedHub

Dashboard

Inventory

Sales & Billing

Suppliers

Sales & Billing

localhost:5173 says

Enter quantity:

12

OKCancel

Name	Category	Price	Quantity	Add
Paracetamol	Analgesic			
Amoxicillin	Antibiotic	₹25.00	48	Add
Cetirizine	Antihistamine	₹15.00	75	Add
Ibuprofen	Anti-inflammatory	₹12.00	80	Add
Omeprazole	Antacid	₹20.00	60	Add

Inventory

Medicine Name	Category	Quantity	Expiry	Price
Paracetamol	Analgesic	47	2025-12-31T18:30:00.000Z	₹10.50
Amoxicillin	Antibiotic	48	2025-12-30T18:30:00.000Z	₹25.00
Cetirizine	Antihistamine	75	2026-06-29T18:30:00.000Z	₹15.00
Ibuprofen	Anti-inflammatory	80	2025-10-14T18:30:00.000Z	₹12.00
Omeprazole	Antacid	55	2026-03-19T18:30:00.000Z	₹20.00

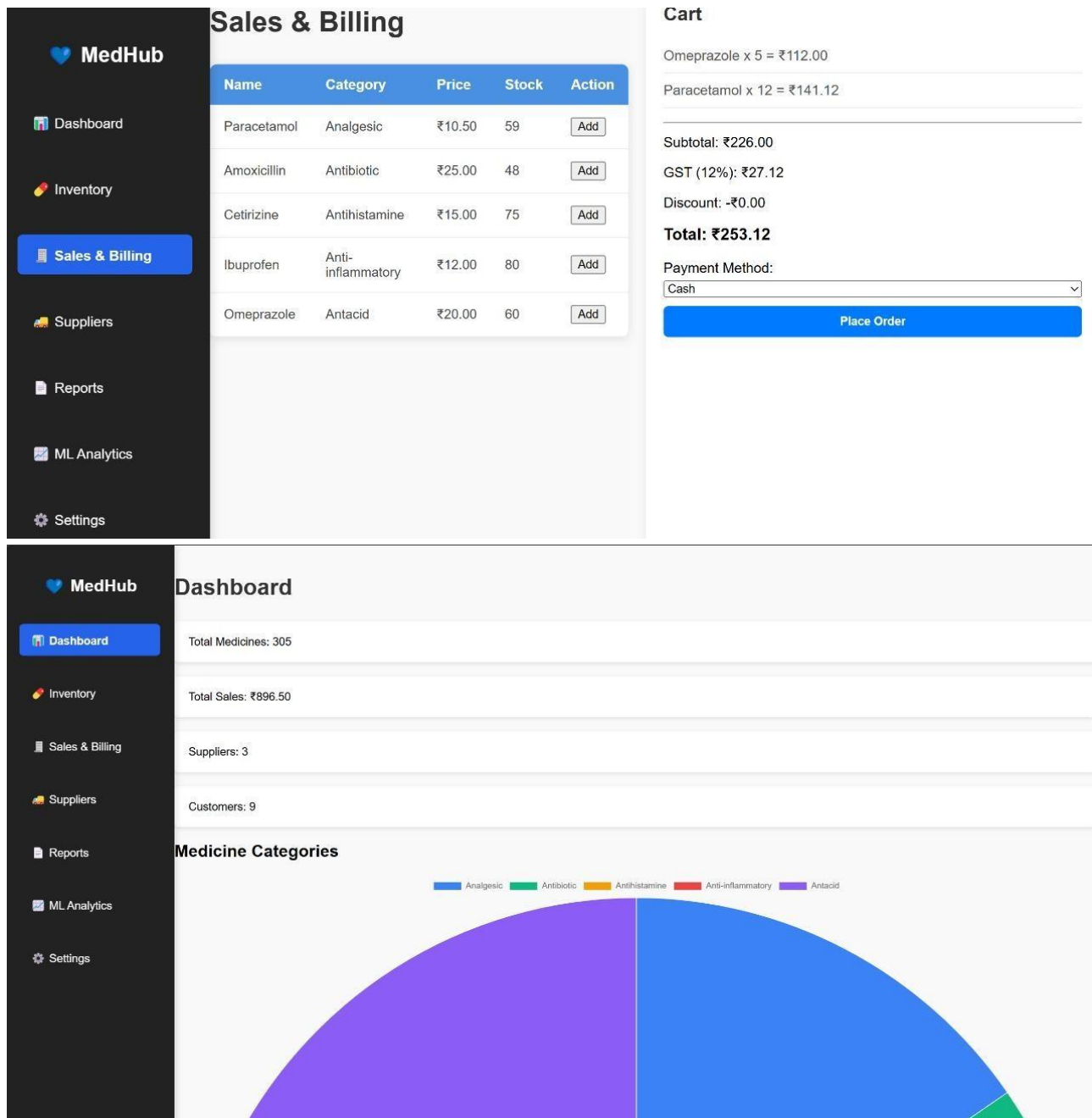


Figure 6.1: MEDHUB System Dashboard Overview

Chapter 7

Conclusion and Future Prospects

7.1 Summary of Achievements The MEDHUB system successfully:

- Implements a fully normalized relational model for pharmacy data.
- Tracks medicine usage, prescriptions, and supplier orders.
- Supports prediction-based medicine recommendation.
- Enhances accountability through role-based access and reports.

7.2 Limitations

- Limited visualization of long-term trends.
- Basic front-end suitable for academic demonstration.
- No integration with hospital systems or IoT devices in current version.
- Machine learning modules and reporting dashboards are not fully implemented.

7.3 Future Enhancements

- Real-time integration with IoT-based stock monitoring.
- Enhanced dashboard for historical trend analysis and custom reporting.
- AI-driven recommendation engine for medicine alternatives.
- Integration with hospital EHR systems for prescription validation.
- Mobile version for customer-side medicine ordering.
- Full integration of machine learning models for disease forecasting.
- Visual report generation module for sales, demand, and stock health.

7.4 Lessons Learned

- Proper normalization simplifies complex data operations.
- Cross-table relationships improve performance and accuracy.
- Data integrity through foreign key enforcement is critical.
- Early inclusion of predictive models enhances usability.

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